

Why Trading Speed Matters: A Tale of Queue Rationing under Price Controls

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We show that queue rationing under price controls is one driver of high-frequency trading. Uniform tick sizes constrain price competition and create rents for liquidity provision, particularly for securities with lower prices. The time priority rule allocates rents to high-frequency traders (HFTs) because of their speed advantage. An increase in relative tick size, defined as uniform tick sizes divided by security prices, increases the fraction of liquidity provided by HFTs but harms liquidity. We find that the message-to-trade ratio is a poor cross-sectional proxy for HFTs' liquidity provision: stocks with more liquidity provided by HFTs have lower message-to-trade ratios. (*JEL* G10, G20)

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When regulations prevent a price from reaching its market-clearing level, the divergence between the marginal valuation of a good and its price creates economic rents (Suen 1989). How people spend resources to capture the rents depends on the allocation rule applied in that market. The “first come, first served” rule generates queuing, or early arrival to the market to beat rivals (Kornai 1980; Shleifer and Vishny 1991, 1992). In this paper, we show that

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queuing helps explain high-frequency trading, in which the competition for front queue positions under constrained price competition leads to an arms race in trading speed.

Stock exchanges in the United States are organized as electronic limit order books. A trader can act as a liquidity provider by posting a limit order for a specified price and quantity. A trade occurs when a liquidity demander accepts a limit order by submitting a market order. Limit orders are first ranked by price: sell limit orders at lower prices and buy limit orders at higher prices are executed ahead of those with less aggressive prices. In practice, price competition among liquidity providers is constrained by the 1-cent uniform tick size rule, particularly for stocks with lower prices.¹ For limit orders queuing at the same price, the time priority rule gives precedence to the order arriving first. Tick size constraints and the time priority rule amount to price control and queue rationing, essentially generating a queuing channel of speed competition.

We first provide evidence in support of the queuing channel using a stratified sample of 117 Russell 3000 stocks in 2010. The 1-cent uniform tick size implies that low-priced stocks have higher relative tick sizes. We show that tick size creates rents for liquidity provision. The revenue margins for liquidity provision are higher for stocks with larger relative tick sizes. The higher rents, in turn, lengthen the queue to provide liquidity at the best bid and offer (BBO). Next, we show that the time priority rule allocates the rents created by tick size to high-frequency traders (HFTs), who use relatively more limit orders as relative tick sizes increase, consistent with the increased revenue margin of liquidity provision. Surprisingly, the difficulty with establishing the time priority of limit orders compels non-HFTs to submit more market orders as the relative tick size increases, even though a larger relative tick size increases the revenue from submitting limit orders. Taking rent creation and rent allocation together, a larger relative tick size increases the liquidity provided by HFTs relative to the liquidity provided by non-HFTs.

As a robustness check, we use splits/reverse splits of leveraged exchange-traded funds (ETFs) as exogenous shocks on relative tick sizes. In our difference-in-differences tests, the treatment group includes leveraged ETFs that split/reverse split, while the control group includes leveraged ETFs that track the same index but do not split/reverse split. We find that splits increase the proportional quoted spread, depth at the BBO, and the fraction of liquidity provided by HFTs. These findings are consistent with the queuing channel: stock splits widen the relative tick size, constrain price competition, and increase the proportional quoted spread. Liquidity providers quoting

¹ The Securities and Exchange Commission's (SEC's) Rule 612 in Regulation NMS prohibits stock exchanges from displaying, ranking, or accepting quotations for, orders for, or indications of interest in any NMS stock priced in an increment smaller than \$0.01 if the quotation, order, or indication of interest is priced equal to or greater than \$1.00 per share.

heterogeneous prices before splits may quote identical prices afterwards, which lengthens the queue and helps HFTs to secure time priority; the opposite occurs with reverse splits.

The literature on speed competition in liquidity provision predominately focuses on the speed of order *cancellation* to avoid the risk of being adversely selected (see Jones 2013 and Menkveld 2016 for surveys). Liquidity providers post bid or ask quotes at which they will buy or sell shares of an asset. When new information arrives, their quotes become stale. If they have a speed advantage, HFTs can quickly cancel stale quotes before they are adversely selected. The focus on order cancellations in the previous literature leads to three predictions. First, HFTs incur lower adverse selection costs. As a consequence, they can post more competitive quotes and essentially drive slower liquidity providers out of the market (Bernales 2014; Han, Khapko, and Kyle 2014; Hoffmann 2014; Bongaerts and Van Achter 2016). Second, HFTs are more likely to crowd out slow liquidity providers when tick size is smaller, because a smaller tick size reduces the constraints to offer better prices (Chordia et al. 2013). Third, because of the continuous updating process for new information, high-frequency liquidity providers tend to submit and cancel a large number of orders for each transaction (Jones 2013), and an increase in speed leads to more quote updates between trades (Menkveld 2016). Therefore, researchers who do not have account-level data on HFTs often use the message-to-trade ratio as a proxy for HFTs' activities, especially for their liquidity provision activity (see Biais and Foucault (2014) for a survey).

We contribute to the literature by showing a new dimension of speed competition: quick order *submission* to achieve time priority. This paper shows three results that oppose predictions focusing on order cancellation, but support theoretical predictions of the queuing channel in Wang and Ye (2017). First, we find that an increase in the risk of adverse selection crowds out HFTs' liquidity provision. Wang and Ye (2017) provide one possible interpretation for this result. When adverse selection risk is low, tick size can be wider than the unconstrained bid-ask spread. HFTs are able to crowd out non-HFTs through time priority in the queue. An increase in adverse selection risk widens unconstrained bid-ask spread. In turn, tick size constraints are less binding and non-HFTs are able to win execution priority through price. Second, we find that a large relative tick size crowds out the liquidity provision of non-HFTs, particularly when the nominal bid-ask spread is binding at one tick. This result is also consistent with Wang and Ye (2017), who show that non-HFTs have incentives to establish price priority over HFTs when tick sizes are small. Third, we find that the message-to-trade ratio is a poor cross-sectional proxy for HFTs' liquidity provision. Stocks with higher fractions of liquidity provided by HFTs have lower message-to-trade ratios. Wang and Ye (2017) rationalizes this surprising result. HFTs provide more liquidity for stocks with larger tick sizes, but they have less incentive to cancel orders after achieving top queue positions; HFTs provide less liquidity for stocks with smaller tick

sizes, but they cancel orders more frequently because price competition occurs on a finer grid.

Our paper is closely related to Budish, Cramton, and Shim (2015), who find that continuous-time trading creates adverse selection rents and drives high-frequency trading.² We show that discrete pricing also drives high-frequency trading. Rents in the queuing channel arise from tick size rather than adverse selection, but both types of rents originate from market design and lead to an arms race in speed.

The queuing channel casts doubt on the recent policy proposal in the United States to increase the tick size, initiated by the 2012 Jumpstart Our Business Startups Act (the JOBS Act). In October 2016, the SEC started a two-year pilot program to increase the tick size to 5 cents for 1,200 less liquid stocks. Proponents to increase the tick size assert that a larger tick size should control the growth of HFTs and increase liquidity. Our results indicate that an increase in tick size would *encourage* HFTs and increase effective spread, or the actual transaction cost for liquidity demanders.

1. Sample Construction and Data

In this section, we describe our two samples of securities. In Section 1.1, we describe a NASDAQ HFT data set with 117 randomly selected stocks. The HFT data set reports aggregated quotes and trades for 26 firms identified as HFTs. Quotes and trades from all other traders are identified as non-HFTs. We use the NASDAQ HFT data set to examine the cross-sectional variation in HFTs' liquidity provision. In Section 1.2, we describe the treatment and control groups for our ETF splits/reverse splits sample. We use Bloomberg to create our sample of ETFs, and we use NASDAQ TotalView-ITCH (ITCH for short) and TAQ data to calculate HFT activities.

1.1 Stock sample and NASDAQ HFT data

Our first sample includes 117 stocks from the NASDAQ HFT data set. When the sample was originally selected in early 2010, it included 40 large stocks from the 1000 largest Russell 3000 stocks, 40 medium stocks ranked from 1001 to 2000, and 40 small stocks ranked from 2001 to 3000. Three of these 120 stocks were delisted in October 2010. Panel A of Table 1 contains the summary statistics for the 117 stocks in October 2010.

We use three types of files from the NASDAQ HFT data set: the trade data set, the minute-by-minute limit order book snapshots, and the data set on quote updates. The trade data set contains each trade in NASDAQ excluding trades occurring in the opening, closing, and intraday crosses. Each trade record includes ticker symbol, price, number of shares, timestamp in milliseconds,

² Biais, Foucault, and Moinas (2015) and Foucault, Kozhan, and Tham (2017) also consider the rents created by adverse selection.

Table 1
Summary statistics

	Mean	SD	Min.	Median	Max.	Obs.
A. NASDAQ HFT October sample						
<i>tick_{relative}</i>	0.050	0.039	0.002	0.037	0.192	2,457
<i>logmcap</i>	21.913	1.891	19.371	21.410	26.399	2,457
<i>DolDep</i> (in million)	0.092	0.217	0.002	0.018	1.868	2,457
<i>pHFTvolume</i>	0.282	0.137	0.000	0.268	0.728	2,457
<i>pHFTlimit</i>	0.459	0.159	0.000	0.416	0.875	2,457
<i>pNonHFTlimit</i>	0.803	0.106	0.406	0.819	1	2,457
<i>logbv_{average}</i>	13.124	2.105	8.822	13.196	17.881	2,310
<i>idiorisk</i>	0.013	0.019	0.001	0.008	0.139	2,310
<i>age</i> (in 1k days)	9.664	7.799	0.945	7.565	30.955	2,310
<i>numAnalyst</i>	13.845	10.145	1	11.5	50	2,310
<i>PIN</i>	0.118	0.052	0.021	0.111	0.275	2,310
<i>RevenueMargin</i> (in bps)						
<i>1-min Interval</i>	0.719	3.903	−80.319	0.246	88.889	4,914
<i>5-min Interval</i>	0.552	4.489	−39.360	0.244	145.615	4,914
<i>30-min Interval</i>	0.574	6.777	−69.213	0.401	145.615	4,914
<i>Daily Closing</i>	1.657	18.741	−323.384	1.372	215.426	4,914
B. NASDAQ HFT February sample						
<i>tick_{relative}</i>	0.054	0.042	0.002	0.038	0.217	585
<i>logmcap</i>	21.818	1.929	19.562	21.322	25.947	585
<i>pTimePriority</i>	0.330	0.246	0.000	0.245	0.918	585
<i>pHFTTimePriority</i>	0.420	0.153	0.000	0.419	1.000	585
<i>pNonHFTundercut</i>	0.573	0.209	0.047	0.571	1.000	585
<i>pOnecentTime</i>	0.414	0.379	0.000	0.231	0.995	585
<i>pOnecentVol</i>	0.415	0.322	0.000	0.305	0.979	585
<i>logbv_{average}</i>	13.124	2.106	8.822	13.196	17.881	550
<i>idiorisk</i>	0.014	0.020	0.001	0.008	0.160	550
<i>age</i> (in 1k days)	9.442	7.804	0.723	7.343	30.733	550
<i>numAnalyst</i>	13.273	9.999	1	10	43	550
<i>PIN</i>	0.109	0.066	0.000	0.091	0.377	550
C. LETF split/reverse split sample						
<i>RelativeRun</i>	0.221	0.041	0.026	0.221	0.382	1,280
<i>DolDep</i> (in million)	0.235	0.629	0.009	0.095	6.994	1,280
<i>pQspread</i> (in bps)	10.420	5.715	1.775	9.590	33.166	1,280
<i>pQspread_{cum}</i> (in bps)	12.797	6.926	2.443	11.627	44.668	1,280
<i>pEspread</i> (in bps)	7.692	4.893	1.505	6.853	67.894	1,280
<i>pEspread_{cum}</i> (in bps)	10.110	6.436	2.163	8.717	70.870	1,280
<i>return</i>	0.000	0.041	−0.217	−0.001	0.205	1,280

This table reports summary statistics. Panel A reports the results of stocks in the NASDAQ HFT sample in October 2010; panel B reports the results of stocks in the NASDAQ HFT sample for February 22–26, 2010; and panel C reports the results of the leveraged ETF sample for the difference-in-differences test, in which the event window is 5 days before splits/reverse splits from 2010 through 2013. The appendix provides definitions of the variables. All the variables are measured for each stock day except for RevenueMargin, which contains two observations for each stock day, one for HFTs and one for non-HFTs.

and buy/sell indicator referring to the liquidity-seeking side of the trade. NASDAQ categorizes executions into four types: “HH,” HFTs take liquidity from other HFTs; “HN,” HFTs take liquidity from non-HFTs; “NH,” non-HFTs take liquidity from HFTs; and “NN,” non-HFTs take liquidity from other non-HFTs.³ NASDAQ defines liquidity provision as the use of nonmarketable limit

³ Non-HFTs in NASDAQ HFT data set also can be sophisticated traders. Hasbrouck and Saar (2013) discuss the distinction between proprietary algorithms (HFTs) and agency algorithms, which are used by buy-side institutions

orders and liquidity demand as the use of market orders.⁴ A trader can switch from providing liquidity to demanding liquidity by changing the order type. The limit order book snapshot file contains 391 1-minute snapshots of the NASDAQ book from 9:30 a.m. to 4:00 p.m. EST for each trading day in the sample period.

The trade file and the snapshots of the limit order book do not contain the queue positions of the orders. Fortunately, the NASDAQ also provides a data set that contains each update to the price or size of HFTs' and non-HFTs' best quotes. Although the data set we employ is for a short sample period (February 22–26, 2010), the focus of our paper is on the cross-sectional variation of HFTs' activity, rather than their time-series variation. Panel B of Table 1 reports the summary statistics for the 117-stock sample for the February 22–26, 2010, period.

When a stock has a 1-cent bid-ask spread, the nominal tick size is binding and a liquidity provider can no longer undercut existing limit orders on NASDAQ. The summary statistics of two variables in panel B of Table 1 are worth mentioning: *pOnecentTime* and *pOnecentVol*. These two variables measure the magnitude of the binding nominal tick size. The quote-based measure, *pOnecentTime*, is the fraction of time a stock has a 1-cent bid-ask spread within a trading day. The trade-based measure, *pOnecentVol*, is the fraction of volume executed when the bid-ask spread is 1 cent. On average, a stock in our sample has a 1-cent bid-ask spread 41.4% of the time, and 41.5% of trading volume occurs when the bid-ask spread is exactly 1 cent.

The NASDAQ HFT data suffer from two limitations, which are unlikely to affect our results. First, the NASDAQ cannot identify all HFTs.⁵ Including some HFTs in the non-HFTs group tends to bias the estimate of their differences toward zero. Still, we find economically and statistically significant activity differences between HFTs and non-HFTs that demonstrate the robustness of our results. Second, the NASDAQ HFT data set does not include trading activity in other exchanges. To establish price priority over standing limit orders on the NASDAQ, a trader can submit a limit order at the same price to an exchange that charges lower fees for market orders but higher fees for limit orders (Yao and Ye 2014; Battalio, Corwin, and Jennings 2016; Chao, Yao, and Ye 2017a). The exclusion of other exchanges, however, is unlikely to affect the cross-sectional variation in tick size constraints because of two reasons: (1) SEC Rule 612 applies to all stock exchanges, so the cost to establish price priority is higher

to minimize the cost of executing trades in the process of implementing changes in their investment portfolios. These agency algorithms are slower than HFTs (Hasbrouck and Saar 2013) and show up as non-HFTs in the NASDAQ HFT data set.

⁴ Technically, market orders in NASDAQ are marketable limit orders, while limit orders are nonmarketable limit orders. We use market orders and limit orders for short.

⁵ High-frequency trading desks in large and integrated firms (e.g., Goldman Sachs and Morgan Stanley) may be excluded because these institutions also act as brokers for customers and engage in proprietary low-frequency strategies, so their orders cannot be uniquely identified as high-frequency or non-high-frequency business. The other omission involves orders from small HFTs that route their orders through these integrated firms (Brogaard, Hendershott, and Riordan 2014).

for low-priced stocks in any exchange, and (2) for stocks priced above \$1, the fees of given exchanges do not vary with the stock price, thus the cost of establishing price priority across exchanges is also higher for low-priced stocks.⁶

1.2 Sample of leveraged ETFs

In the ETF splits/reverse splits experiment, we use leveraged ETFs that undergo a split/reverse split as the treatment group and use leveraged ETFs that track the same indexes but do not split/reverse split in the event window as the control group.⁷ Leveraged ETFs are ETFs that seek to deliver multiples of the performance of the index or benchmark they track. They often appear in pairs to track the same index in opposite directions. For example, SPXL amplifies S&P 500 returns by 300%, whereas SPXS amplifies S&P 500 returns by -300% . These twin-leveraged ETFs usually share the same issue prices, and their issuers frequently use splits/reverse splits to align their nominal prices after initial public offerings (IPOs).

We identify leveraged ETFs using Bloomberg and collect the dates of the splits/reverse splits from CRSP. Our event windows include 5 days before and 5 days after the splits/reverse splits. If an ETF splits/reverse splits multiple times during the sample period, we consider each split or reverse split as a separate event. We require leveraged ETFs in our sample to trade each day in the event window and to have at least 50 averaged daily trades. This requirement leaves us with a sample of 17 splits and 47 reverse splits from January 2010, through December 2013. Reverse splits occur more frequently, as ETF issuers are often concerned about the higher trading cost of low-priced ETFs.⁸ Panel C of Table 1 presents the summary statistics of the splits/reverse splits sample.

Leveraged ETF splits/reverse splits provide clean identification for the causal effect of the relative tick size on HFTs' activity, but we are not able to directly measure HFTs' activity, because the NASDAQ HFT data set provides information on HFTs' activity only for 117 stocks. Therefore, we calculate three widely used proxies for HFTs. We use TAQ data to calculate the quote-to-trade ratio (Angel, Harris, and Spatt 2011, 2015) and the negative dollar volume divided by total number of messages (Hendershott, Jones, and Menkveld 2011; Boehmer, Fong, and Wu 2015), and we use ITCH to calculate strategic runs (Hasbrouck and Saar 2013).

⁶ For example, in October 2010, the highest rebates for limit orders were 0.295 cents per share in NASDAQ, whereas liquidity providers pay 0.03 cents per share for limit orders on the Boston Stock Exchange. Thus, the cost of establishing price priority by posting an order at the same price on the Boston Stock Exchange is 0.325 cents for stocks priced above \$1. In percentage terms, the cost is higher for lower-priced stocks.

⁷ We exclude the cases in which one ETF in the pair splits and the other ETF in the pair reverse splits on the same day.

⁸ "See http://www.proshares.com/resources/reverse_split_faqs.html.

2. HFTs' Liquidity Provision Increases with Relative Tick Size

In Sections 2–4, we examine the queuing channel using cross-sectional variation of the relative tick sizes of the 117 stocks. Specifically, in Section 2, we show that an increase in relative tick size increases the fraction of liquidity provided by HFTs, and in Sections 3 and 4, we show how tick size constraints and queue rationing drive this result. Unless otherwise noted, the econometric specification in Sections 2–4 is Equation (1):

$$DepVar_{i,t} = \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + u_{j,t} + \epsilon_{i,t}. \quad (1)$$

DepVar is the dependent variable and the subscript i, t denotes stock i on date t . The key variable of interest, $tick_{relative}$, is the relative tick size (i.e., 1 cent divided by the price). To control for omitted variable bias, we search the literature for control variables (X) that can potentially correlate with nominal price. Benartzi et al. (2009) argue that few variables can explain nominal price, and propose a norms hypothesis with only two explanatory variables: market capitalization and industry. We control for market capitalization and industry-by-time fixed effects ($u_{j,t}$) in our regression. We also take five lines of study in the nominal price literature into consideration, three of which suggest additional control variables in our study.⁹ Panel B of the appendix presents the control variables suggested by these three hypotheses.

Table 2 shows that an increase in relative tick size increases the fraction of trading volume with HFTs as liquidity providers, $pHFTvolume$. For example, Column 1 shows that an increase in the relative tick size from 0.01 to 0.1 (i.e., a decrease in the stock price from \$100 to \$10) is associated with an 8.9% (0.994×0.09) increase in $pHFTvolume$, representing a 31.7% increase relative to its mean. Column 2 shows similar result with additional control variables.

It is surprising that an increase in relative tick size increases HFTs' liquidity provision, because the existing channels on speed competition imply the opposite. Hendershott, Jones, and Menkveld (2011) and Hoffmann (2014) find that the speed advantage of HFTs reduces their adverse selection risk; Brogaard et al. (2015) show that an increase in speed facilitates inventory management; and Carrion (2013) argues that HFTs have lower order processing costs. These reduced intermediation costs should allow HFTs to quote better prices than non-HFTs. Because the constraints to offer strictly better prices are less binding with smaller relative tick size, a reduction in the relative tick size should increase, or at least not decrease, the liquidity provided by HFTs. Consequently, Chordia et al. (2013, p. 644) raise the concern that "HFTs use their speed advantage to crowd out liquidity provision when the tick size is small and stepping in front of standing limit orders is inexpensive."

⁹ Two other lines of research do not indicate additional control variables for our study. Baker, Greenwood, and Wurgler (2009) find time-series variations in stock prices: firms split when investors place higher valuations on low-priced firms and vice versa, but our analysis focuses on cross-sectional variation. Campbell, Hilscher, and Szilagyi (2008) find that an extremely low price forecasts distress risk, but the 117 firms in our sample are far from default.

Table 2
Relative tick size and fraction of liquidity provided by HFTs

Dependent variable	<i>pHFTvolume</i>	
	(1)	(2)
<i>tick_{relative}</i>	0.994*** (0.36)	1.013*** (0.36)
<i>logmcap</i>	0.057*** (0.01)	0.038*** (0.01)
<i>logbv_{average}</i>		0.004 (0.00)
<i>idiorisk</i>		−0.424 (0.46)
<i>age</i>		0.002 (0.00)
<i>numAnalyst</i>		0.001 (0.00)
<i>pin</i>		−0.468** (0.22)
R ²	0.614	0.659
N	2,457	2,310
Industry × Time FEs	Y	Y

This table presents the regression results of the fraction of trading volume with HFTs as liquidity providers on the relative tick size. The regression uses the NASDAQ HFT sample in October 2010. The regression specification is

$$pHFTvolume_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript *i,t* denotes stock *i* on date *t*. *pHFTvolume* is the fraction of volume with HFTs as the liquidity providers. *tick_{relative}* is the relative tick size. *u_{j,t}* is industry-by-time fixed effects for industry *j* on date *t*. The appendix provides definitions of the control variables (*X*). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

We find, however, that HFTs crowd out non-HFTs’ liquidity provision when relative tick size is large, a result suggesting the existence of additional economic drivers of speed competition. We propose a queuing channel of speed competition to fill this gap: a large relative tick size constrains price competition and encourages speed competition to win time priority in the liquidity provision queue. We elaborate the queuing channel in Sections 3 and 4 and provide robustness checks for the queuing channel in Section 5.

Column 2 of Table 2 shows that a one standard deviation increase in adverse selection risk measured by probability of informed trading (PIN) (Easley et al. 1996) decreases the fraction of liquidity provided by HFTs by 2.4% (0.052*0.468), another result inconsistent with the existing economic channels. If a speed advantage allows HFTs to reduce adverse selection costs, they should have a comparative advantage of liquidity provision for stocks with higher adverse selection risk. Wang and Ye (2017) provides one possible explanation for the decrease in liquidity provision by HFTs when adverse selection risk increases. In Wang and Ye (2017), an increase in adverse selection risk increases the unconstrained bid-ask spread relative to the tick size. As a result, an

increase in adverse selection risk encourages price competition, discourages speed competition, and reduces HFTs' liquidity provision.

3. Tick Size Creates Rents and Generates Queues of Liquidity Provision

The first driver of the queuing channel is the tick size constraints, which create rents and generate queues for liquidity provision. In this section, we examine the constrained price competition using cross-sectional variation in relative tick size. We find that a larger relative tick size creates higher revenue for liquidity provision and lengthens the queue to provide liquidity at the BBO.

Following Menkveld (2013), Brogaard, Hendershott, and Riordan (2014), and Baron, Brogaard, Hagströmer, and Kirilenko (forthcoming), we define HFTs' liquidity provision revenue for a stock throughout a time interval as

$$\pi_{HFT} = \sum_{k=1}^K CASH_{HFT}^k + INV_{HFT} \times P_{mid}, \quad (2)$$

where k denotes transaction k , and K denotes the total number of transactions in the interval, $CASH_{HFT}^k$ denotes the cash flow for each of the HFTs' transactions, and $\sum_{k=1}^K CASH_{HFT}^k$ captures their total cash flows throughout the interval.¹⁰ $INV_{HFT} \times P_{mid}$ represents value changes in net position in each interval by clearing the inventory cumulated in the interval (INV_{HFT}) at the end of the interval midpoint quote (P_{mid}). We consider four interval lengths: 1 minute, 5 minutes, 30 minutes, and 1 day.¹¹

We denote L as the length of an interval and V as the total number of intervals in a day.¹² The daily revenue margin for HFTs with inventory clearance frequency L is

$$RevenueMargin_{HFT}^L = \sum_{v=1}^V \frac{\pi_{HFT,v}}{DolVol_{HFT}}, \quad (3)$$

where $DolVol_{HFT}$ is the total daily dollar volume with HFTs as the liquidity providers. The revenue margins for non-HFTs are calculated analogously.

We then run the following regression:

$$\begin{aligned} RevenueMargin_{i,t,n}^L = & \beta_1 \times HFTdummy_{i,t,n} + \beta_2 \times Dmtick_{relative_{i,t}} \\ & + \beta_3 \times HFTdummy_{i,t,n} \times Dmtick_{relative_{i,t}} + u_{j,t} + \Gamma \times X_{i,t} \\ & + \epsilon_{i,t,n}, \end{aligned} \quad (4)$$

¹⁰ We include liquidity rebate in the first part of the profit. The NASDAQ has a complex fee structure; we use a rebate of 0.295 cents per share in calculating profit, but the results are similar at other rebate levels.

¹¹ Brogaard, Hendershott, and Riordan (2014) assume that inventory is cleared daily at the closing midpoint. Evidence shows that the inventories of HFTs cross zero multiple times a day. For example, Brogaard et al. (2015) find that the inventory of HFTs can cross zero 13.32 times a day.

¹² Each trading day contains 6.5 hours. If the interval L is 30 minutes, the total number of intervals V equals 13.

Table 3
Relative tick size and liquidity provision revenue

Dependent Variable	Revenue margin (in bps)							
	1-min interval		5-min interval		30-min interval		Daily closing	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Dmtick_{relative}</i>	15.438*** (5.08)	20.188*** (5.42)	13.830*** (3.92)	17.943*** (4.57)	12.980*** (4.90)	14.463*** (4.75)	24.315 (16.09)	33.637* (17.75)
<i>HFTdummy</i>	0.823*** (0.16)	0.878*** (0.14)	0.823*** (0.19)	0.878*** (0.18)	0.384* (0.22)	0.471*** (0.18)	-0.202 (0.52)	-0.143 (0.54)
<i>Dmtick_{relative}</i> × <i>HFTdummy</i>	7.313 (5.25)	6.070 (5.60)	12.119 (7.89)	11.614 (8.62)	9.661 (6.72)	5.311 (6.67)	-4.768 (17.78)	-13.563 (18.09)
<i>logmcap</i>	-0.154** (0.07)	-0.176 (0.20)	-0.074 (0.07)	-0.012 (0.22)	-0.041 (0.07)	-0.027 (0.14)	-0.167 (0.19)	0.218 (0.49)
<i>logbvaverage</i>		0.053 (0.08)		0.092 (0.11)		0.030 (0.07)		0.342* (0.20)
<i>idiorisk</i>		-7.236 (8.21)		-4.874 (9.33)		3.661 (5.71)		0.378 (20.16)
<i>age</i>		0.001 (0.02)		-0.012 (0.02)		-0.022 (0.02)		-0.038 (0.06)
<i>numAnalyst</i>		0.027 (0.02)		0.024 (0.02)		0.031* (0.02)		0.014 (0.06)
<i>pin</i>		2.864 (4.06)		4.153 (4.83)		2.690 (3.05)		12.824 (9.25)
R ²	0.206	0.226	0.195	0.215	0.186	0.198	0.168	0.179
<i>N</i>	4,914	4,620	4,914	4,620	4,914	4,620	4,914	4,620
Industry × Time FEs	Y	Y	Y	Y	Y	Y	Y	Y

This table presents the regression results of the revenue margin of liquidity provision on the relative tick size. The regression uses the NASDAQ HFT sample in October 2010. The regression specification is

$$\begin{aligned} \text{RevenueMargin}_{i,t,n}^L = & \beta_1 \times \text{Dmtick}_{relative,i,t} + \beta_2 \times \text{HFTdummy}_{i,t,n} \\ & + \beta_3 \times \text{Dmtick}_{relative,i,t} \times \text{HFTdummy}_{i,t,n} + u_{j,t} + \Gamma \times X_{i,t} + \epsilon_{i,t,n}, \end{aligned}$$

where subscript i, t denotes stock i on date t . RevenueMargin_n^L is the daily revenue margin assuming inventory cleared at frequency L of trader type n . L is taken to be 1 minute for Columns 1 and 2, 5 minutes for Columns 3 and 4, 30 minutes for Columns 5 and 6, and daily closing for Columns 7 and 8. Trader type n takes two values: HFTs and non-HFTs. $\text{Dmtick}_{relative}$ equals relative tick size minus its sample mean. HFTdummy_n equals 1 if the revenue measure is for HFTs, and zero if the revenue measure is for non-HFTs. $u_{j,t}$ is industry-by-time fixed effects for industry j on date t . The appendix provides definitions of the control variables (X). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

where $n \in \{\text{HFTs, non-HFTs}\}$. HFTdummy_n equals 1 for HFTs' revenue margin and zero for non-HFTs' revenue margin. $\text{Dmtick}_{relative}$ is $\text{tick}_{relative}$ minus its sample mean.¹³ $u_{j,t}$ captures the industry-by-time fixed effects for industry j on date t . X represents the control variables in panel B of the appendix.

Table 3 shows that the revenue margin of liquidity provision increases with the relative tick size. For example, Column 2 shows that an increase in the relative tick size from 0.01 to 0.1 increases the revenue margin with 1-minute inventory clearance by 1.82 bps (20.188*0.09). Results for

¹³ Without demeaning, β_1 captures the difference between HFTs' and non-HFTs' revenue margins for stocks with zero tick size (infinitely high price).

other clearance frequencies are qualitatively similar, although Columns 7 and 8 show statistically weaker results for the revenue margin with daily inventory clearance. The weaker results imply the value of higher frequency inventory clearance. Indeed, one important feature of HFTs is their “very short time-frames for establishing and liquidating positions” (SEC 2010, p. 45).

The increase of liquidity provision revenue with respect to the relative tick size we document is consistent with the results of Anshuman and Kalay (1998) and O’Hara, Saar, and Zhong (2015), who argue that HFTs’ higher revenue margins for larger relative tick size stocks can explain their desire to be more actively engaged in liquidity provision for those stocks. Table 3, however, shows that an increase in relative tick size leads to a statistically identical increase in HFTs’ and non-HFTs’ revenue margin, which implies that the increased rents led by an increase in relative tick size provides similar incentives for both types of traders to increase liquidity provision. To explain the increase of HFT liquidity provision relative to non-HFT liquidity provision, we need to consider both rent creation and rent allocation. In the next section, we show that the time priority rule allows HFTs to capture a larger fraction of rents created as a result of an increase in relative tick size.

An increase in relative tick size also lengthens the queue for liquidity provision at the BBO. We measure the queue length using $DolDep_{i,t}$, the time-weighted dollar depth at the NASDAQ BBO for stock i on day t .¹⁴ We regress $DolDep_{i,t}$ on relative tick size and control variables following Equation (1). Table 4 shows that the dollar depth increases with relative tick size. Column 1 shows that an increase in the relative tick size from 0.01 to 0.1 increases the dollar depth by \$0.18 million (2.049×0.09).

4. Speed Allocates Rents in the Queue

The second driver of the queuing channel, queue rationing, allocates rents created by tick size to traders with higher speed. In this section, we explore queue rationing using cross-sectional variation of the relative tick sizes of the 117 stocks. In Section 4.1, we demonstrate that, as relative tick size increases, HFTs use more limit orders to provide liquidity. Surprisingly, we find that non-HFTs use more market orders to demand liquidity despite an increase in revenue margin led by an increase in relative tick sizes. In the rest of Section 4, we rationalize the opposite trends of liquidity supply and demand between HFTs and non-HFTs. In Section 4.2, we show that a *large* relative tick size enables HFTs to establish time priority. In Section 4.3, we show that a *small* relative tick size helps non-HFTs to establish price priority. We provide further discussion on time versus price priority in Section 4.4.

¹⁴ The time-weighted dollar depth at NASDAQ BBO comes from the snapshots of the limit order book.

Table 4
Relative tick size and dollar depth

Dependent variable	DolDep	
	(1)	(2)
<i>tick_{relative}</i>	2.049*** (0.63)	2.430*** (0.73)
<i>logmcap</i>	0.083*** (0.02)	0.075*** (0.03)
<i>logbvaverage</i>		0.001 (0.01)
<i>idiorisk</i>		−0.722 (0.60)
<i>age</i>		0.004 (0.00)
<i>numAnalyst</i>		0.002 (0.01)
<i>pin</i>		0.241 (0.42)
R ²	0.570	0.596
N	2,457	2,310
Industry × Time FEs	Y	Y

This table presents the regression results of dollar depth at NASDAQ BBO on the relative tick size. The regression uses the NASDAQ HFT sample in October 2010. The regression specification is

$$DolDep_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript *i,t* denotes stock *i* on date *t*. *DolDep* is the dollar depth at NASDAQ BBO. *tick_{relative}* is the relative tick size. *u_{j,t}* is industry-by-time fixed effects for industry *j* on date *t*. The appendix provides definitions of the control variables (*X*). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

4.1 HFTs’ and non-HFTs’ choice of market versus limit orders

In this subsection, we examine whether an increase in relative tick size incentivizes HFTs to use more limit orders to supply liquidity.

We measure HFTs’ fraction of volume from providing liquidity, *pHFTlimit*, as (*NH* + *HH*) divided by (*HN* + *HH* + *NH*) for each stock day. Analogously, we measure non-HFTs’ fraction of volume from providing liquidity, *pNonHFTlimit*, as (*HN* + *NN*) divided by (*NH* + *NN* + *HN*) for each stock day.

Table 5 shows panel regression results following Equation (1). Columns 1 and 2 show that *pHFTlimit*, HFTs’ fraction of trading volume from providing liquidity, increases with relative tick size. This result is consistent with the increased revenue of liquidity provision led by a larger relative tick size. Surprisingly, Columns 3 and 4 show that *pNonHFTlimit* decreases with relative tick size, suggesting that non-HFTs use less limit orders but more market orders as the relative tick size increases. For example, Column 4 shows that an increase in the relative tick size from 0.01 to 0.1 decreases non-HFTs’ fraction of trading volume from supplying liquidity by 10% (1.116*0.09). The order choice of

Table 5
Relative tick size and the fraction of limit orders

Dependent variable	<i>pHFTlimit</i>		<i>pNonHFTlimit</i>	
	(1)	(2)	(3)	(4)
<i>tick_{relative}</i>	1.649*** (0.41)	1.869*** (0.42)	-1.040*** (0.31)	-1.116*** (0.32)
<i>logmcap</i>	0.054*** (0.01)	0.032** (0.01)	-0.041*** (0.00)	-0.030*** (0.01)
<i>logbv_{average}</i>		0.002 (0.01)		-0.004 (0.00)
<i>idiorisk</i>		-0.957** (0.46)		0.391 (0.37)
<i>age</i>		0.002 (0.00)		-0.002 (0.00)
<i>numAnalyst</i>		0.002 (0.00)		-0.000 (0.00)
<i>pin</i>		-0.460 (0.28)		0.292 (0.19)
R ²	0.489	0.545	0.539	0.584
N	2,457	2,310	2,457	2,310
Industry × Time FEs	Y	Y	Y	Y

This table presents the regression results of the fraction of volume from using limit orders on the relative tick size. The regression uses the NASDAQ HFT sample for October 2010. The regression specification is

$$DepVar_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript i,t denotes stock i on date t . In Columns 1 and 2, $DepVar$ represents $pHFTlimit$, HFTs' fraction of volume from using limit orders. In Columns 3 and 4, $DepVar$ represents $pNonHFTlimit$, non-HFTs' fraction of volume from using limit orders. $tick_{relative}$ is the relative tick size. $u_{j,t}$ is industry-by-time fixed effects for industry j on date t . The appendix provides definitions of the control variables (X). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

non-HFTs not only departs from that of HFTs but also contradicts with the increased revenue for liquidity provision.

Wang and Ye (2017) provides one economic mechanism to reconcile these contradictions: an increase in relative tick size increases liquidity provision revenue, but it also lengthens the liquidity provision queue. Non-HFTs, which do not have a speed advantage in obtaining the front queue position, are forced to demand liquidity to achieve order executions. We elaborate this economic mechanism in Sections 4.2–4.4.

4.2 A large relative tick size enables HFTs to establish time priority

In this subsection, we examine whether a large relative tick size helps HFTs to achieve time priority over non-HFTs. To the best of our knowledge, we are the first to differentiate order executions due to liquidity providers' *time* priority from those due to liquidity providers' *price* priority.

We start our analysis by linking each trade to its corresponding quote update in the limit order book.¹⁵ If, at the time of execution, both HFTs and non-HFTs

¹⁵ We match the trade data set and the quote update data set by their millisecond timestamp, sign, price, size of the order (trade), and the type of liquidity provider (HFTs or non-HFTs).

Table 6
Relative tick size and dollar volume executed through time priority

Dependent variable	<i>pTimePriority</i>		<i>pHFTTimePriority</i>	
	(1)	(2)	(3)	(4)
<i>tick_{relative}</i>	3.326*** (0.47)	3.754*** (0.63)	0.986*** (0.24)	1.164*** (0.35)
<i>logmcap</i>	0.106*** (0.01)	0.058*** (0.02)	0.030*** (0.01)	0.026*** (0.01)
<i>logbvaverage</i>		0.003 (0.01)		−0.002 (0.00)
<i>idiorisk</i>		−1.408 (1.48)		−0.622 (0.45)
<i>age</i>		0.007*** (0.00)		0.001 (0.00)
<i>numAnalyst</i>		0.007* (0.00)		−0.002 (0.00)
<i>pin</i>		−0.220 (0.33)		−0.545** (0.24)
R ²	0.707	0.753	0.357	0.441
N	585	550	585	550
Industry × Time FEs	Y	Y	Y	Y

This table presents the regression results of the fraction of dollar volume due to limit orders having time priority on the relative tick size. The regression uses the NASDAQ HFT sample for February 22–26, 2010. The regression specification is

$$DepVar_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript i,t denotes stock i on date t . In Columns 1 and 2, *DepVar* represents *pTimePriority*, the dollar volume executed through time priority over total dollar volume. In Columns 3 and 4, *DepVar* represents *pHFTTimePriority*, HFTs' dollar volume executed through time priority divided by the total dollar volume executed through time priority. *tick_{relative}* is the relative tick size. $u_{j,t}$ is industry-by-time fixed effects for industry j on date t . The appendix provides definitions of the control variables (X). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

provide quotes at the execution price, we classify this execution as a *time* priority trade. If, at the time of execution, only one type of trader provides quotes at the execution price, we classify this execution as a *price* priority trade. We use this method to classify trades into four types: (1) liquidity-providing HFTs have time priority; (2) liquidity-providing non-HFTs have time priority; (3) liquidity-providing HFTs have price priority; and (4) liquidity-providing non-HFTs have price priority.

Table 6 reports the regression results based on Equation (1). The sample period is February 22–26, 2010. The dependent variable in Columns 1 and 2 is *pTimePriority*, the dollar volume executed through time priority relative to total dollar volume, calculated as the dollar volume from type 1 and type 2 trades over the dollar volume from all four types of trades. We find that time priority becomes more important as relative tick size increases. For example, Column 2 shows that *pTimePriority* increases by 33.8% (3.754×0.09) when relative tick size increases from 0.01 to 0.1. These results suggest that queue rationing becomes more important for stocks with larger relative tick sizes.

In Columns 3 and 4 of Table 6, the dependent variable is *pHFTTimePriority*, the dollar volume from HFTs' orders winning time priority relative to total dollar volume from orders winning time priority. We measure

pHFTTimePriority as the dollar volume from type 1 trades divided by the sum of the dollar volume from type 1 and type 2 trades. We find that a rise in relative tick size leads to an increase in *pHFTTimePriority*. For example, Column 4 shows that an increase in relative tick size from 0.01 to 0.1 increases *pHFTTimePriority* by 10.5% (1.164×0.09). This result suggests that HFTs are more likely to establish time priority over non-HFTs as relative tick size increases.

In summary, a larger relative tick size increases the fraction of liquidity provided by HFTs through two mechanisms: it elevates the fractions of volumes executed through time priority, and it increases the likelihood that HFTs will establish time priority over non-HFTs.

4.3 A small relative tick size helps non-HFTs establish price priority

Chordia et al. (2013) raise the concern that HFTs may be more likely to undercut non-HFTs when the relative tick size is small. In the queuing channel, however, non-HFTs have greater incentives to establish price priority, because they are less likely to establish time priority over HFTs. We use NASDAQ limit order book update data for February 22–26, 2010, to examine whether a decrease in relative tick size incentivizes non-HFTs to quote better prices than HFTs.

We start by checking, for each quote update, whether the best bid price after the top-of-book update is higher than the previous best bid price, or whether the best ask price after the top-of-book update is lower than the previous ask price. If so, we regard this update as an undercutting order. *pNonHFTUndercut* equals the aggregate dollar sizes of non-HFTs' undercutting orders divided by the total dollar sizes of the undercutting orders.

Table 7 reports the regression results based on Equation (1). We find that non-HFTs' fraction of the undercutting dollar size increases as relative tick size declines. Column 1 shows that *pNonHFTUndercut* increases by 11% (1.217×0.09) when relative tick size decreases from 0.1 to 0.01. This result suggests that a small relative tick size incentivizes non-HFTs to undercut existing limit orders and establishes price priority.

4.4 The relative tick size and the binding nominal tick size

In this subsection, we examine whether stocks with larger relative tick sizes are more likely to have a binding nominal tick size, for which liquidity providers are unable to further undercut existing limit orders. Such a pattern would provide a rationalization for the increased use of market orders by non-HFTs as relative tick size increases.

Table 8 reports the regression results based on Equation (1). An increase in relative tick size dramatically increases the probability of a binding nominal tick size. For example, Column 1 shows that the fraction of time with a 1-cent bid-ask spread rises by 48.6% (5.4×0.09) as relative tick size increases from 0.01 to 0.1. The same increase in relative tick size raises the fraction of trading volume executed when the bid-ask spread is 1 cent by 43.8% (4.87×0.09) (Column 4).

Table 7
Relative tick size and undercutting orders

Dependent variable	<i>pNonHFTundercut</i>	
	(1)	(2)
<i>tick_{relative}</i>	-1.217** (0.48)	-1.026* (0.60)
<i>logmcap</i>	-0.058*** (0.01)	-0.043* (0.02)
<i>logbv_{average}</i>		-0.015 (0.01)
<i>idiorisk</i>		-0.079 (0.82)
<i>age</i>		-0.003 (0.00)
<i>numAnalyst</i>		0.001 (0.00)
<i>pin</i>		0.544 (0.36)
R ²	0.459	0.484
<i>N</i>	585	550
Industry × Time FEs	Y	Y

This table presents the regression results of the fraction of dollar volume that improves BBO from non-HFTs' orders on the relative tick size. The regression uses the NASDAQ HFT sample for February 22-26, 2010. The regression specification is

$$pNonHFTundercut_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript i,t denotes stock i on date t . $pNonHFTundercut$ is non-HFTs' fraction of the aggregate dollar size of orders that improve the BBO. $tick_{relative}$ is the relative tick size. $u_{j,t}$ is industry-by-time fixed effects for industry j on date t . The appendix provides definitions of the control variables (X). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

For traders who do not have a speed advantage, undercutting existing limit orders and using market orders are two alternatives to achieve execution in the NASDAQ limit order book. An increase in relative tick size increases the likelihood of a binding nominal tick size, which eliminates the option of undercutting existing limit orders. This potential episode provides an explanation of why non-HFTs use more market orders despite an increase in revenue through using limit orders. Although a larger relative tick size increases the revenue of liquidity provision, it also increases the probability of binding nominal tick size, thereby forcing non-HFTs to demand liquidity.

5. Robustness Checks Using ETF Splits/Reverse Splits

As robustness checks, we use difference-in-differences tests to examine further whether a large relative tick size increases the liquidity provided by HFTs. In Section 5.1, we discuss a race among three proxies for HFTs' liquidity provision. In Section 5.2, we present the difference-in-differences test using

Table 8
Relative tick size and binding nominal tick size

Dependent variable	<i>pOnecentTime</i>		<i>pOnecentVol</i>	
	(1)	(2)	(3)	(4)
<i>tick_{relative}</i>	5.400*** (0.64)	5.788*** (0.84)	4.462*** (0.56)	4.870*** (0.75)
<i>logmcap</i>	0.149*** (0.02)	0.067** (0.03)	0.128*** (0.02)	0.056** (0.02)
<i>logbverage</i>		0.005 (0.02)		0.001 (0.01)
<i>idiorisk</i>		-1.395 (2.27)		-1.525 (2.01)
<i>age</i>		0.014*** (0.00)		0.011*** (0.00)
<i>numAnalyst</i>		0.012* (0.01)		0.011** (0.01)
<i>pin</i>		-0.277 (0.53)		-0.295 (0.45)
R ²	0.651	0.699	0.660	0.712
N	585	550	585	550
Industry × Time FEs	Y	Y	Y	Y

This table shows the relation between the binding 1-cent nominal tick size and the relative tick size. The regression uses the NASDAQ HFT sample for February 22–26, 2010. The regression specification is

$$DepVar_{i,t} = u_{j,t} + \beta \times tick_{relative_{i,t}} + \Gamma \times X_{i,t} + \epsilon_{i,t},$$

where subscript i, t denotes stock i on date t . In Columns 1 and 2, $DepVar$ represents $pOnecentTime$, the fraction of time that the quoted spread is 1 cent. In Columns 3 and 4, $DepVar$ represents $pOnecentVol$, the fraction of volume from orders executed when the bid-ask spread is 1 cent. $tick_{relative}$ is the relative tick size. $u_{j,t}$ represents industry-by-time fixed effects for industry j on date t . The appendix provides definitions of the control variables (X). We report estimated coefficients and heteroscedasticity robust standard errors, clustered by stock. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

the splits/reverse splits of leveraged ETFs as exogenous shocks to relative tick sizes.

5.1 Measure of HFTs' activity and liquidity

As the NASDAQ HFT data set does not contain any ETFs, our analysis in Section 5 relies on proxies for HFTs' activity. Most studies in the literature on HFTs use some variation of message-to-trade ratio as proxies for HFTs' activity (Boehmer, Fong, and Wu 2015). We use TAQ data to construct two such proxies: the quote-to-trade ratio (Angel, Harris, and Spatt 2011, 2015) and the negative dollar volume divided by total number of messages (Hendershott, Jones, and Menkveld 2011; Boehmer, Fong, and Wu 2015).¹⁶ Biais and Foucault (2014) maintain that the message-to-trade ratio serves as a better proxy for liquidity-providing HFTs than for liquidity-demanding HFTs, because higher message-to-trade ratio implies more order cancellations, a feature of liquidity-providing

¹⁶ Hendershott, Jones, and Menkveld (2011) use the negative dollar volume divided by total number of messages as a proxy for algorithmic trading, a precondition for high-frequency trading. Boehmer, Fong, and Wu (2015) use this measure as a proxy for algorithmic trading, high-frequency trading, and low-latency trading, as they use these three terms interchangeably.

Table 9
Correlation test for the proportion of HFT activity proxy

HFT Measures	<i>pHFTvolume (making)</i>	<i>pHFTvolume (taking)</i>
A. Pearson correlation		
<i>RelativeRun</i>	0.837***	0.377***
	<.0001	<.0001
<i>Quote/Trade ratio</i>	-0.390***	-0.082
	<.0001	0.378
<i>-Trading vol. (in \$100)/Message ratio</i>	-0.252***	-0.274***
	0.006	0.003
B. Spearman correlation		
<i>RelativeRun</i>	0.835***	0.361***
	<.0001	<.0001
<i>Quote/Trade ratio</i>	-0.589***	-0.154*
	<.0001	0.0969
<i>-Trading vol. (in \$100)/Message ratio</i>	-0.554***	-0.506***
	<.0001	<.0001

This table presents the cross-sectional correlations between the proxies for the percentage of liquidity provided by HFTs and the HFT activity calculated from the NASDAQ HFT data set for October 2010. The HFT activity measures include the percentage of volume with HFTs as liquidity providers, *pHFTvolume (making)*, and the percentage of volume with HFTs as liquidity takers, *pHFTvolume (taking)*. The proxy for the percentage of liquidity provided by HFTs includes *RelativeRun*, *Quote-to-trade ratio*, and the *Dollar volume (in \$100)-to-message ratio* multiplied by -1. *p*-values based on 117 cross-sectional stock observations are shown under correlation coefficients; *, **, and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

HFTs. Indeed, Hagströmer and Nordén (2013) find that HFTs who use arbitrage and directional strategies have a similar message-to-trade ratio as non-HFTs, but liquidity-providing HFTs’ message-to-trade ratio is higher than non-HFTs’.

The third proxy for the fraction of liquidity provided by HFTs is *RelativeRun*. To construct the measure, we follow Hasbrouck and Saar (2013) to calculate strategic runs, which link a series of submissions, cancellations, and executions that are likely to form an algorithmic strategy. Strategic runs capture both cancellation and queuing activity. As HFTs’ strategy involves frequent cancellations, a strategic run must contain more than 10 cancellations; each cancellation is followed by a quick resubmission (within 100 milliseconds). Hasbrouck and Saar (2013) use the total time span of all strategic runs for a stock as the proxy for HFTs’ activity, which captures the persistence to stay in the queue. As we focus on HFTs’ liquidity-providing activity relative to the total liquidity-providing activity, we normalize strategic runs by trading volume.¹⁷ The normalized variable, *RelativeRun*, is equal to the log value of the total time span of all strategic runs divided by the log value of dollar volume in the NASDAQ market.¹⁸

We construct the true measure of HFT’s activity and the three proxies for HFTs’ activity for the 117 stocks. Table 9 displays the cross-sectional correlations between the true measure and the proxies. The table shows that

¹⁷ Both the quote-to-trade ratio and the negative dollar volume divided by total number of messages are variables that are normalized by trading activity.

¹⁸ We use the log form so that the results are less sensitive to outliers (Wooldridge 2006).

HFTs' liquidity provision is lower for stocks with higher message-to-trade ratios. This result is surprising, because the SEC (2010) names high message-to-trade ratios as one of the main features of HFTs, and Angel, Harris, and Spatt (2011, 2015) and Boehmer, Fong, and Wu (2015) find that the emergence of HFTs is associated with higher message-to-trade ratios. Despite positive *time-series* correlation between HFTs' activity and message-to-trade ratios documented in the literature, Table 9 shows that message-to-trade ratios are poor *cross-sectional* proxies for HFTs' activity.

Wang and Ye (2017) provides a theoretical interpretation for the negative cross-sectional correlation between message-to-trade ratios and the fraction of liquidity provided by HFTs. An escalation in tick size increases the value to win time priority in the queue, thereby enlarging the fraction of liquidity provided by HFTs. Meanwhile, HFTs are less likely to cancel orders because a larger tick size increases the value to stay in the queue. A decrease in relative tick size reduces the liquidity provided by HFTs, but HFTs more frequently cancel orders because price competition occurs on a finer grid.

Our findings and the theoretical prediction in Wang and Ye (2017) echo Manoj Narang, Tradeworx CEO, who states, "[T]he wider the trading increment, the more size starts aggregating at each price level and the more ties you have in prices as there are fewer prices to be at. This elevates the importance of time priority [which] becomes relative to price priority thereby making the role of speed more important. In addition, wider trading increments also reduce cancellation rates as the fair price of a security can now fluctuate in a wider band" (Serrao et al. 2014, p. 6).

Table 9 shows that *RelativeRun* has a high cross-sectional correlation with *pHFTvolume (making)*, the direct measure for the fraction of liquidity provided by HFTs for the 117 stocks; the Pearson correlation between these two variables is 0.837, and the Spearman correlation is 0.835. One possibility for such high correlation is that *RelativeRun* captures not only fast responses and frequent cancellations but also the persistent interest in supplying liquidity in the queue. Because of these high cross-sectional correlations, we use *RelativeRun* as a proxy for the liquidity provided by HFTs in the ETF difference-in-differences test.

Table 9 shows that *RelativeRun* has a positive, but much lower, correlation with the proportion of liquidity taken by HFTs, because *RelativeRun* mainly captures the persistence of the limit orders in the queue, a factor that is less important for liquidity-demanding activities.¹⁹ Therefore, we use *RelativeRun*

¹⁹ Pure submissions of market orders are not considered strategic runs, because all "runs" start with limit orders. The 10-message cutoff and the time weight also increase the correlation of *RelativeRun* with patient liquidity-providing algorithms. Impatient liquidity-demanding algorithms may use limit orders, but these algorithms are more likely to switch to market orders once the initial limit orders fail to be executed. Therefore, strategic runs that arise from liquidity-demanding algorithms should contain fewer messages. Even if they contain more than 10 messages, it is natural to expect that they span a shorter period of time and carry a lower time weight in *RelativeRun*.

as a proxy for the proportion of liquidity provided by HFTs, although it may capture HFTs' liquidity-taking activity to a small degree. Table 9 also shows that the quote-to-trade ratio and the negative dollar volume divided by total number of messages are negatively correlated with proportion of liquidity taken by HFTs.

To examine the impact of relative tick size on liquidity, we construct three liquidity measures using the NASDAQ ITCH data: (1) the time-weighted daily proportional quoted bid-ask spread ($pQspread$), which is the quoted spread divided by the mid-quote; (2) the size-weighted daily proportional effective spread ($pEspread$), which is the twice the average absolute difference between each transaction price and the mid-quote at the time of the transaction divided by the mid-quote; and (3) the time-weighted daily dollar depth at BBO ($DolDep$). Since a liquidity provider obtains a rebate and a liquidity demander pays a fee for each executed share, we also calculate the cum fee proportional time-weighted quoted bid-ask spread ($pQspread_{cum}$), that is, the sum of the proportional quoted spread and twice the make fee divided by the mid-quote, as well as the cum fee proportional size-weighted effective spread ($pEspread_{cum}$), that is, the sum of the proportional effective spread and twice the take fee divided by the mid-quote.²⁰

5.2 Difference-in-differences tests using leveraged ETF splits

In this subsection, we use difference-in-differences tests to examine the causal relationship between the relative tick size, market liquidity, and the liquidity provided by HFTs. The regression specification is

$$DepVar_{i,t,j} = u_{i,t} + \gamma_{i,j} + \rho \times treatment_{i,j} \times after_{i,t,j} + \theta \times return_{i,t,j} + \epsilon_{i,t,j}, \quad (5)$$

where the subscripts i, t, j identify ETF j in index i at day t . The dummy variable *treatment* equals 1 for the treatment group and zero for the control group. In panel A, the treatment group includes the ETFs that undergo reverse splits, and the control group includes the ETFs that track the same index but do not reverse split around the event days. In panel B, the treatment group includes the ETFs that undergo splits, and the control group includes the ETFs that track the same index but do not split around the event days. The dummy variable *after* equals 1 after the splits/reverse splits and zero before splits/reverse splits. The variable *return* denotes the contemporaneous return for an ETF.

To derive an unbiased estimate of the treatment effect ρ , the treatment must be uncorrelated with the error term. As we control for both index-by-time fixed effects and ETF fixed effects in Equation (5), the estimation of ρ is biased only if the actual split/reverse split is related to the contemporaneous idiosyncratic

²⁰ We set the liquidity maker's rebate at 0.295 cents per share like in Section 2, and we set the take fee at 0.3 cents per share.

shocks to the fraction of liquidity provided by HFTs.²¹ Two stylized facts mitigate the concern about contemporaneous idiosyncratic shocks.

First, the motivation for ETF splits/reverse splits is simple and transparent. The issuers of ETFs split an ETF when it has a drastically higher nominal price than its pair or reverse split an ETF when it has a drastically lower nominal price than its pair. The ETF fixed effects capture their price differences pre-splits/reverse splits. Second, the schedules for splits/reverse splits is predetermined and announced. Also, fund companies often conduct multiple splits/reverse splits on the same day for ETFs tracking diverse underlying assets.²² The pre-determined schedule and the diversified sample further mitigate the concern that splits/reverse splits decisions may correlate with contemporaneous idiosyncratic shocks.

Column 1 in panel A of Table 1 shows that reverse splits reduce the fraction of liquidity provided by HFTs, which is consistent with our predictions for the queuing channel. The coefficient of -0.012 represents a 5.4% decrease relative to the mean (0.221, in panel C of Table 1). Columns 2–4 provide further evidence to support the queuing channel. Columns 2 and 3 show that the proportional quoted spread decreases, implying that a reduction in relative tick size elevates price competition in liquidity provision.²³ A more intense price competition also leads to a shorter queue. Indeed, Column 4 shows that dollar depth drops by \$304,000. In summary, reverse splits create finer proportional price grids, which provide an environment that incentivizes price competition, discourages queuing at the same price, and reduces the fraction of liquidity provided by HFTs.

As reverse splits lead to a decrease in proportional quoted spread as well as a reduction in dollar depth, we further examine the effective spread, the most relevant measure of transaction costs incurred by liquidity demanders (Bessembinder 2003). Columns 5 and 6 in Table 10 show that reverse splits lead to a decrease in the proportional effective spread by 4.80 bps, along with a decrease in cum fee proportional effective spread by 8.65 bps. These results suggest that reverse splits reduce transaction costs for liquidity demanders.

Panel B of Table 10 shows the pattern for splits. Column 1 demonstrates that HFTs' liquidity provision increases; Columns 2 and 3 show that proportional quoted spread widens; Column 4 reports that dollar depth at BBO, or the length of the queue, increases after splits. All three results support the queuing channel. A coarse price grid constrains price competition, thereby increasing the proportional quoted spread. A coarse price grid can also force traders who

²¹ We adopt a similar identification strategy to Hendershott, Jones, and Menkveld (2011), and we also follow their discussion on endogeneity.

²² For example, the announcement on April 9, 2010, involves leveraged ETFs for oil, gas, gold, real estate, financial stocks, basic materials, and Chinese indices.

²³ The cum fee proportional quoted spread decreases more than ex fee proportional quoted spread, because reverse splits also reduce rebates proportionately.

Table 10
Difference-in-differences test using leveraged ETF splits (reverse splits)

Dependent variable	<i>RelativeRun</i> (1)	<i>pQspread</i> (in bps) (2)	<i>pQspread_{cum}</i> (in bps) (3)	<i>DolDep</i> (in mn) (4)	<i>pEspread</i> (in bps) (5)	<i>pEspread_{cum}</i> (in bps) (6)
A. Reverse split sample						
<i>treatment</i> × <i>after</i>	−0.012** (0.01)	−4.166*** (0.66)	−7.952*** (1.13)	−0.304** (0.14)	−4.802*** (0.75)	−8.652*** (1.21)
<i>return</i>	−0.053** (0.02)	−6.215*** (1.81)	−9.420*** (2.78)	0.599 (0.53)	−1.440 (2.83)	−4.700 (3.92)
R ²	0.861	0.920	0.880	0.722	0.796	0.796
<i>N</i>	940	940	940	940	940	940
Index × Time FE	Y	Y	Y	Y	Y	Y
ETF FEs	Y	Y	Y	Y	Y	Y
B. Split sample						
<i>treatment</i> × <i>after</i>	0.009** (0.00)	2.060*** (0.69)	3.696*** (1.03)	0.086 (0.07)	2.486*** (0.72)	4.150*** (1.17)
<i>return</i>	−0.002 (0.01)	−10.488*** (2.45)	−12.275*** (2.59)	−0.059 (0.07)	−7.156*** (1.84)	−8.973*** (2.21)
R ²	0.872	0.923	0.918	0.867	0.911	0.909
<i>N</i>	340	340	340	340	340	340
Index × Time FEs	Y	Y	Y	Y	Y	Y
ETF FEs	Y	Y	Y	Y	Y	Y

This table presents results for difference-in-differences tests using leveraged ETF splits (and reverse splits) from 2010 to 2013. The event windows are 5 days before and 5 days after the splits/reverse splits. The regression specification is

$$DepVar_{i,t,j} = u_{i,t} + \gamma_{i,j} + \rho \times treatment_{i,j} \times after_{i,t,j} + \theta \times return_{i,t,j} + \epsilon_{i,t,j},$$

where subscript i, t, j denotes ETF j of index i on date t . *DepVar* represents *RelativeRun*, the proxy for the fraction of liquidity provided by HFTs in Column 1; *pQspread*, the time-weighted proportional quoted spread in Column 2; *pQspread_{cum}*, the cumulative fee time-weighted proportional quoted spread in Column 3; *DolDep*, the time-weighted dollar depth at the NASDAQ BBO in millions of dollars in Column 4; *pEspread*, the size-weighted proportional effective spread in Column 5; and *pEspread_{cum}*, the cumulative fee size-weighted proportional effective spread in Column 6. $u_{i,t}$ is the index-by-time fixed effects for index i on date t , and $\gamma_{i,j}$ denotes the ETF fixed effects for ETF j of index i . In panel A (B), the treatment dummy, *treatment*, equals 1 for ETFs that undergo reverse split (split) and zero for ETFs that do not undergo reverse split (split) in the index pair. The dummy variable *after* equals zero before splits/reverse splits and 1 after the splits/reverse splits. *Return* denotes the contemporaneous return for the ETF. We report estimated coefficients and heteroscedasticity robust standard errors, clustered by ETF. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

quote heterogeneous bid-ask prices before splits to quote identical prices after splits, thereby possibly lengthening the queue at the best quote. The longer queue then incentivizes speed competition to achieve time priority. Columns 5 and 6 show that splits increase the proportional effective spread, the transaction costs for the liquidity demanders.

In summary, our robustness checks show that an increase in relative tick size increases HFTs’ liquidity provision. An increase in relative tick size also increases proportional quoted spread, dollar depth, and the transaction costs for liquidity demanders. One innovation of our empirical design is that the treatment and control groups share identical fundamentals. This approach not only helps us to isolate the queuing channel from other interpretations of speed competition but also allows us to contribute to the literature on securities splits.

Berk and DeMarzo (2013) report mixed results in the literature about the impact of stock splits on liquidity. Our paper shows that splits increase transaction costs compared with securities with identical fundamentals that do not split.²⁴

6. Conclusion

In this paper, we demonstrate a queuing channel for speed competition in liquidity provision. Tick size creates rents for liquidity provision, and the time priority rule distributes the rents to traders with higher speed capabilities. We find that a large relative tick size increases the liquidity provided by HFTs, but the allocation is not due to price competition. Instead, a large relative tick size constrains price competition, leads to queue rationing, and favors traders with ultrahigh trading speed. The difficulty of establishing time priority compels non-HFTs to submit more market orders as the relative tick size increases, even though a larger relative tick size increases the revenue from submitting limit orders. An increase in relative tick size following ETF splits encourages speed competition among traders and increases transaction costs; a decrease in relative tick size following ETF reverse splits discourages speed competition and decreases transaction costs.

We find that non-HFTs are more likely to establish price priority in the limit order book than are HFTs when the relative tick size is small, but we do not know the identity of these non-HFTs. These traders may be fundamental investors who use limit orders to minimize the transaction costs of their portfolio adjustment. Frazzini, Israel, and Moskowitz (2014) find that a large fundamental institutional investor designs trading algorithms to provide rather than demand liquidity. Wang and Ye (2017) find that the incentive to complete a trade allows natural buyers and sellers to undercut HFTs as long as aggressive limit orders are less costly than market orders. A promising future direction of research is to examine the identity and the incentives of the non-HFTs who undercut the price of HFTs, using more detailed data on trader types.

The 2012 JOBS Act aims to jumpstart capital formation by increasing tick size. Our results cast doubt on the effectiveness of this initiative. We find that instead of generating revenue to support sell-side equity research and to increase the number of IPOs, a large tick size would reduce liquidity, create more rents that would drive the speed competition to achieve the top position in the queue, and fuel another round of the arms race in trading speed. We encourage policy makers to consider *decreasing* the tick size, particularly for large stocks with lower prices.

High-frequency trading is not the only response to a discrete tick size. Chao, Yao, and Ye (2017a,b) show that a discrete tick size also leads to proliferations of U.S. stock exchanges. The literature on market microstructure focuses on

²⁴ Muscarella and Vetsuypens (1996) examine seven solo ADR splits, but there is no control group in their liquidity results.

liquidity and price discovery given market structures, but the formation of a market structure is also endogenous. We believe that a new and fruitful line of research is to examine why certain market structures exist in the first place.

Appendix

Table A1
Variable descriptions

A. Variable of interest	Description
<i>tick_{relative}</i>	Relative tick size (1 divided by stock price in dollars)
<i>Dmtick_{relative}</i>	Relative tick size minus its sample mean (0.05)
<i>DolDep</i> (in million)	Time-weighted dollar depth at NASDAQ BBO
<i>pHFTvolume</i>	Trading volume with HFTs as liquidity providers divided by total trading volume
<i>pHFTlimit</i>	Trading volume with HFTs as liquidity providers divided by total trading volume with HFTs as either liquidity providers or demanders
<i>pNonHFTlimit</i>	Trading volume with non-HFTs as liquidity providers divided by total trading volume with non-HFTs as either liquidity providers or demanders
<i>pTimePriority</i>	Dollar volume due to liquidity providers gaining time priority divided by total dollar volume
<i>pHFTTimePriority</i>	Dollar volume due to HFTs gaining time priority divided by the total dollar volume due to liquidity providers gaining time priority
<i>pNonHFTundercut</i>	Non-HFTs' aggregate dollar size of orders that improve the BBO divided by the total dollar size of orders that improve the BBO
<i>pOnecentTime</i>	Time span that the quoted spread is 1 cent divided by the length of regular trading hours
<i>pOnecentVol</i>	Trading volume executed when the quoted spread is 1 cent divided by total trading volume
<i>RevenueMargin</i> (in bps)	
<i>1-min Interval</i>	Revenue margin assuming minute-by-minute inventory clearance at 1-minute midpoint
<i>5-min Interval</i>	Revenue margin assuming inventory clearance every 5 minutes at the 5-minute midpoint
<i>30-min Interval</i>	Revenue margin assuming inventory clearance every 30 minutes at the 30-minute midpoint
<i>Daily Closing</i>	Revenue margin assuming daily inventory clearance at daily closing midpoint
<i>RelativeRun</i>	Proxy for fraction of liquidity provided by HFTs
<i>pQspread</i> (in bps)	Time-weighted proportional quoted spread
<i>pQspread_{cum}</i> (in bps)	Cum fee time-weighted proportional quoted spread
<i>pEspread</i> (in bps)	Size-weighted proportional effective spread
<i>pEspread_{cum}</i> (in bps)	Cum fee size-weighted proportional effective spread
B. Control variables	Description
<i>logmcap</i>	Logarithm value of market capitalization
<i>idiorisk</i>	Variance on the residual from a 60-month beta regression using CRSP Value Weighted Index. Angel (1997) includes <i>idiorisk</i> in the optimal tick size hypothesis, which predicts that firms choose the optimal tick size through splits/reverse splits
<i>numAnalyst</i>	Number of analysts providing one-year earnings forecasts. Angel (1997) includes <i>numAnalyst</i> in the optimal tick size hypothesis, which predicts that firms choose the optimal tick size through splits/reverse splits
<i>age</i> (in 1k days)	Length of time for which price information is available for a firm on the CRSP monthly file. Angel (1997) includes <i>age</i> in the optimal tick size hypothesis, which predicts that firms choose the optimal tick size through splits/reverse splits
<i>logbv_{average}</i>	Logarithm of the average book value of equity per shareholder at the end of the previous year, a proxy for small investor ownership, suggested by Dyl and Elliott (2006). Dyl and Elliott (2006) use <i>logbv_{average}</i> to test the marketability hypothesis that a lower price appeals to individual traders
<i>PIN</i>	Probability of informed trading (Easley et al. 1996). Easley, O'Hara, and Saar (2001) use <i>PIN</i> to test the signaling hypothesis that firms use stock splits to signal good news
<i>return</i>	Contemporaneous daily return

This table contains descriptions of the variables.

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